INCH-POUND

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DETAIL SPECIFICATION

ARMOR PLATE, TITANIUM ALLOY, WELDABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers a weldable wrought titanium alloy structural armor plate in nominal thicknesses from 0.25 - 4.0 inch, inclusive.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This sections does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. ARMY TEST AND EVALUATION COMMAND (USATECOM)

TOP 2-2-710 Ballistic Tests of Armor Materials

ITOP 4-2-805 Projectile Velocity and Time of Flight Measurements

(Application for copies should be addressed to the Defense Technical Information Center, 8725 John J. Kingman Road, Suite 0944, Fort Belvoir, VA 22060-6218.)

Beneficial comments (recommendations, additions, deletions0 and any pertinent data which may be of use in improving this document should be addressed to: Director, U.S. Army Research Laboratory, Weapons and Materials Research Directorate Aberdeen Proving Ground, MD 21005- 5069 by using the self-addressed Standardization Document Improvement Proposal(DD Form 1426) appearing at the end of this document or by letter.

AMSC/NA FSC 9535

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 480	General Requirements for Flat-Rolled Stainless and Heat Resisting Steels in Form of Plate, Sheet, and Strip
ASTM E 8	Tension Testing of Metallic Materials
ASTM E 120	Chemical Analysis of Titanium and Titanium-Base Alloys
ASTM E 539	X-Ray Emission Spectrometric Analysis of 6Al-4V Titanium Alloy
ASTM E 1409	Determination of Oxygen in Titanium and Titanium Alloys by the Inert gas Fusion Technique
ASTM E 1447	Determination of Hydrogen in Titanium and Titanium Alloys by Inert Gas Fusion Thermal Conductivity Method

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, Conshohocken, PA 19428-2959.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE AMS 2249	Chemical Check Analysis Limits - Titanium and Titanium Alloys
SAE ARP 982	Minimizing Stress Corrosion Cracking in Wrought Titanium Alloy Products

(Application for copies should be addressed to the SAE, Publications Group, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ Z1.4	Sampling Procedures and Tables for Inspection by Attributes

(Application for copies should be addressed to the ASQ, 611 E. Wisconsin Ave, Milwaukee, WI 53202-3005.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.4 Order of precedence. In the event of a conflict of between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.1), the contractor shall furnish a sample or samples for first article inspection and approval. The contractor shall comply with this requirement at the time of his first order or contract and at any time that the supplier has not furnished wrought armor in the

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applicable thickness range under this specification within a period of 18 months (see 4.4).

3.2 Chemical composition.

3.2.1 Class composition. The chemical composition of the plates shall be as specified in Table 1 for classes 1, 2 and 3. If no class is specified on the contract, drawing or order, Class 1 shall be supplied. Chemical composition variances shall meet the requirements of AMS 2249.

Table 1. Chemical Composition - Percent by Weight

									Other[3]	
Class	Al	V	C	O	N	H[1]	Fe	Ti	Each	Total
1	5.50 - 6.50	3.50 - 4.50	0.04 max	0.14 max	0.02 max	0.0125 max	0.25 max	Rem[2]	0.10 max	0.40 max
2	5.50 - 6.75	3.50 - 4.50	0.08 max	0.20 max	0.05 max	0.0150 max	0.30 max	Rem[2]	0.10 max	0.40 max
3	5.50 - 6.75	3.50 - 4.50	0.08 max	0.30 max	0.05 max	0.0150 max	0.30 max	Rem[2]	0.10 max	0.40 max
4			0.08 max	0.30 max	0.05 max	0.0150 max		Rem[2]		

[1] Hydrogen shall be determined on each lot of the product as shipped.

[2] Titanium is determined by differences.

[3] Other elements need not be analyzed nor reported unless otherwise specified.

3.2.2 Class 4 composition. Upon approval of the procuring activity, an alternate composition of titanium alloy may be authorized. In addition to the composition limits specified in Table 1, suppliers must submit to the procuring activity established limits for each intentionally added element as well as maximum limits for others.

3.2.3 Class 4 density. The density shall be 0.158 through 0.165 pounds per cubic inch. Density may be calculated based on chemical composition range established.

3.3 Mechanical properties. Plate shall meet the minimum mechanical properties of Table 2 either in the longitudinal or transverse directions, after final heat treatment. After temperature exposure and cooling, a minimum of 0.001 inch shall be removed from all surfaces of the sample prior to testing. Sufficient material shall be removed so as to insure compliance of the sample with the requirements of 3.9.

Table 2. Minimum Mechanical Properties

Class	Yield Strength 0.2% Offset (psi)	Tensile Strength (psi)	Elongation (%)
1	110,000	120,000	10
2	110,000	120,000	6
3	110,000	120,000	6
4	110,000	120,000	6

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3.4 Ballistic requirements. The minimum ballistic limits at 0° obliquity shall be in accordance with the values shown in Tables 7 through 10. No through cracks greater than 2.5 inch in length on any plate surface are permitted after ballistic testing.

3.5 Dimensional tolerances.

3.5.1 Length, width, and thickness. Length and width tolerances shall be as agreed upon by purchaser and vendor. The plus and minus tolerances on plate thickness shall be as specified in Table 3. However, because the specification requirements specified in table 8 end with an ordered thickness 1.570 inch, there is only one range of thickness that will be applicable to table 8 thickness range(1.065 to 1.570 inches). Table 8 can only be ordered to the 1.000 to 1.999 thickness range value below.

Table 3. Plate Thickness Tolerances

Thickness Range (inch)	Tolerance (inch)[1]
0.250 to 0.374	+0.045
0.375 to 0.749	+0.055
0.750 to 0.999	+0.060
1.000 to 1.999	+0.070
2.000 to 2.999	+0.085
3.000 to 4.000	+0.100

[1] The entire range has a minus tolerance of 0.010 inch.

3.5.2 Straightness. Plates shall be of such straightness that the maximum edgewise curvature (depth of arc) shall not exceed 1/8 inch in any 5 ft of length.

3.5.3 Flatness. Flatness tolerances for plates shall conform to Table 4.

Table 4. Flatness Tolerances for Plate

Plate Thickness (inch)	Maximum Permissible Variation in Flatness (inch)								
	Plate Width (inch)								
	Up to 48, incl.	48 to 60, excl.	60 to 72, excl.	72 to 84, excl.	84 to 96, excl.	96 to 108, excl.	108 to 120, excl.	120 to 144, excl.	144 and over
0.250 to 0.374	11/16	3/4	15/16	1-1/8	1-3/8	1-7/16	1-9/16	1-7/8	--
0.375 to 0.499	1/2	9/16	11/16	3/4	15/16	1-1/8	1-1/4	1-7/16	1-3/4
0.500 to 0.749	1/2	9/16	5/8	5/8	13/16	1-1/8	1-1/8	1-1/8	1-3/8
0.750 to 0.999	1/2	9/16	5/8	5/8	3/4	13/16	15/16	1	1-1/8
1.000 to 1.499	1/2	9/16	9/16	9/16	11/16	11/16	11/16	3/4	1
1.500 to 4.000	3/16	5/16	3/8	7/16	1/2	9/16	5/8	3/4	7/8

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3.6 Marking for identification. Unless otherwise specified in the contract or order, each plate shall be marked on one face with the manufacturer's name or trademark, the basic number and revision letter of this specification, composition, the nominal plate thickness, and the lot number. Plate shall be marked in lengthwise rows of characters recurring at intervals not greater than 3 feet, the rows being spaced not more than 6 inches apart and alternately staggered. The characters shall be not less than 3/8 inch in height and shall be applied using a suitable marking fluid whose residue shall contain not more than traces of halogen-bearing compounds and shall be capable of being removed in hot alkaline cleaning solution without rubbing. The markings shall have no deleterious effect on the material or its performance and shall be sufficiently stable to withstand normal handling.

3.6.1 Marking of ballistic test plates. In addition to the marking specified in 3.6, each ballistic test plate shall be marked with the letters PRE if a first article test plate and ACC if an acceptance test plate. When practicable, the marking shall be an impression stamped on the edge of ballistic test plates.

3.7 Heat treatment. Product shall be heat treated within temperature, time, and atmospheric condition ranges established during first article testing or previous production.

3.7.1 Reheating. Unless otherwise approved by the procuring activity, after the final heat treatment, material shall not be reheated or processed above the final annealing, stress relieving, or aging temperature or 100° F below the beta transus, whichever is less. If plates are retreated above that temperature, they shall be recorded and tested as a separate lot.

3.8 Ballistic test plate information. A Titanium Armor Test Data Form (see Figure 1), properly executed, shall be submitted with each ballistic test plate.

3.9 Finish. Unless otherwise specified, plate shall have a surface finish in accordance with ASTM A 480. Product shall be free of surface contamination by oxygen (alpha case), hydrogen, carbon, nitrogen, or other such harmful contaminants. The producer shall be permitted to remove minor surface of the imperfections by spot grinding if such grinding does not reduce the thickness of the material below the minimum permitted by the tolerance for the thickness ordered.

3.10 Workmanship. The material produced under this specification shall be uniform in quality and condition, and free from harmful alloy segregation and surface contamination by oxygen, nitrogen, or other contaminant and foreign material. The plate shall be clean, sound, smooth and free of defects detrimental to fabricability or serviceability such as hard spots, laminations, inclusions, pits, folds, seams, grind marks, and cracks.

4. VERIFICATION

4.1 Clarification of inspections. The inspection requirements specified herein are classified as conformance inspection (see 4.2).

4.2 Conformance inspection. Test for acceptance of individual lots consist of testing all requirements specified in section 3. Verification conformance Inspection is in accordance with the provisions herein.

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4.2.1 Government test activity. Unless otherwise specified, ballistic test plates shall be forwarded for testing to the US Army Aberdeen Test Center, Building 358, Attn: STEAC-SL-V, Aberdeen Proving Ground, MD 21005-5059. Ballistic tests shall be performed at a government test activity selected by the procuring activity (see 6.2).

4.3 Inspection cassifications. Inspection shall be classified as follows:

a. First article inspection (see 4.4).
b. Conformance inspection (acceptance)(see 4.5).

4.3 Lot. A lot shall consist of all plates of the same nominal thickness from the same heat, processed together as a unit. All plates in a lot shall receive the same final heat treatment within standard parameters, not necessarily at the same time.

4.4 Sampling.

4.4.1 For first article inspection.

4.4.1.1 Chemical analysis. A sample for chemical analysis shall be taken from each heat.

4.4.1.2 Mechanical properties. At least two tension test specimens shall be taken from each lot. The location and type of specimens shall be as specified in 4.4.2.2 and 4.4.2.3.

4.4.1.3 Ballistic test samples. Two plates, of the thickness to be supplied on the contract, shall be submitted to a Government approved facility for ballistic tests. One sample shall be taken from the first plate fabricated and one from the last plate fabricated in the initial lot produced. When only one plate is fabricated, the samples shall be taken from opposite ends of the plate. The required plate sizes are listed in Table 5. When more than one thickness is involved, first article approval may be obtained for the size ranges listed in Table 5 by submission of plates of one specified thickness within each range. All ballistic tests shall be conducted at zero degrees obliquity.

4.4.2 Sampling for quality conformance acceptance testing.

4.4.2.1 For chemical analysis. A sample sufficient to perform the analysis required to determine compliance with 3.2 shall be selected from each lot.

Table 5. Ballistic Test Plate Sizes and Corresponding Test Projectiles for First Article and Acceptance Testing

Thickness Ranges (inches)	Minimum Size (inches)	Ballistic Test Projectile
0.250 to 0.615, excl	12 x 12	Cal. .30 AP M2
0.615 to 1.575, excl.	18 x 12	20-mm FSP
1.575 to 1.950, excl	18 x 12	14.5-mm API B32
1.950 to 4.000	18 x 12	20-mm API-T M602

4.4.2.2 For mechanical properties.

4.4.2.2.1 Number of samples. Two random samples shall be taken from each lot. If a lot consists of only one plate, only one sample shall be selected. At least one tension test specimen shall be made from each sample.

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4.4.2.2.2 Types of test specimens. Tension test specimens shall conform to the largest obtainable round specimen shown in ASTM E 8.

4.4.2.2.3 Location of test specimens. Transverse test specimens shall be taken from the plate perpendicular to the primary rolling direction with the central longitudinal axis midway between the top and bottom surfaces of the plate. Longitudinal specimens shall be prepared in such a manner that the longitudinal axis of the specimen is parallel to the principal rolling direction of the sample.

4.4.2.3 For ballistic testing. One plate, 12 in by 12 in for thicknesses less than 0.615 in or 18 in by 12 in for thicknesses 0.615 in and greater, by the thickness of the plate represented, shall be selected from every lot of material of the same thickness. In the event of reduced testing (see 4.6.4), multiple lots represented by a single acceptance ballistic test shall have been manufactured using the same heat treatment and other processing within standard mill parameters of a fixed process.

4.5 Examination.

4.5.1 Visual. Each piece in each lot shall be visually examined for compliance with the requirements for workmanship (see 3.9) and marking for identification (see 3.6).

4.5.2 Dimensions. All plates shall be subject to inspection for compliance with the dimensional and tolerance requirements (see 3.5).

4.5.3 Preparation for delivery. Prior to shipment, examination shall be made to determine compliance with the requirements of section 5.

4.6 Tests (first article and quality conformance).

4.6.1 Chemical composition. Conformance with the chemical composition requirements shall be determined either by heat or lot analysis using ASTM E 539 X-ray emission spectrometric method for aluminum, vanadium and iron. Oxygen shall be analyzed by the inert gas fusion technique specified in ASTM E 1409. Hydrogen shall be analyzed by the inert gas fusion thermal conductivity method specified in ASTM E 1447 and shall be determined on each lot as shipped. Carbon and nitrogen shall be determined using ASTM E 120. Other analytical methods may be used with the written approval of the acquiring activity. In case of a dispute, the results of chemical analysis by the ASTM E 120 method shall be used as the basis for acceptance or rejection.

4.6.2 Mechanical properties. Conformance to the mechanical property requirements shall be determined by testing at least two specimens (one longitudinal and one transverse) in the condition ordered from each sample selected in accordance with 4.4.2.2. tension tests shall be accomplished in accordance with the following provisions.

4.6.2.1 Tension specimen configuration. The tension specimens shall conform to the following configuration in accordance with ASTM E 8:

a. For product thickness up to 0.375 inch, standard "sheet-type" rectangular tension specimens shall be used.

b. For product thickness equal to or greater than 0.375 inch, standard round tension specimens with a 0.250 inch diameter minimum reduced test section shall be used.

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4.6.2.2 Tension specimen preparation. Unless the lot consists of only one unit, not more than one specimen per grain direction per test condition shall be taken from each sample. Longitudinal specimens shall be prepared in such a manner that the longitudinal axis of the specimen is parallel to the principal rolling direction of the sample. Transverse specimens shall be prepared in such a manner that the longitudinal axis of the specimen is perpendicular to the principal rolling direction of the sample.

4.6.2.3 Tension test method. Tension tests shall be accomplished in accordance with ASTM E 8, except that the strain rate shall be 0.003 to 0.007 inch/inch per minute through the yield strength and then is increased so as to produce failure in approximately one additional minute.

4.6.2.4 In case of dispute, the results of referee tension tests performed on a tensile machine having a strain pacer and using a strain rate of 0.005 inch/inch per minute through the yield strength and a minimum crosshead speed of 0.10 inch per minute above the yield strength shall govern.

4.6.3 Ballistic tests. Ballistic test plates shall be tested at an approved facility designated by the procuring activity. Plate thickness as determined by the ballistic test agency shall be used to determine the required protection ballistic limit for the plate. With the exception of table 10, plate thickness shall be taken as the average of 4 thickness measurements read to the nearest 0.001 in and the average reported to the nearest 0.005 in. Table 10 plate thickness measurements are to be read to the nearest 0.01 in, and the average values for the four thickness measurements will be reported to the nearest 0.05 in. At least one measurement will be taken along each edge of the plate at a distance of at least 1 in from the edge. The V_{50} ballistic tests shall be performed in accordance with USATECOM Ballistic Tests of Armors, TOP 2-2-710 and Projectile Velocity and Time of Flight Measurements, ITOP 4-2-805.

4.6.3.1 Cracking. No through cracks greater than 2.5 inch in length on any plate surface are permitted after ballistic testing.

4.6.4 Reduced testing for acceptance. Upon the approval of the procuring activity, reduction in the amount of testing per ASQ Z1.4 may be authorized. This authorization can be granted by the Light Armor Test facility at the Aberdeen Test Center, US TECOM, providing that Ballistic Test Reports show that the results on consecutive lots indicate a uniform product (meeting the requirements) is being produced. Reduction in testing will only be considered if production is steady as determined by the procuring activity. Testing for a given thickness shall return to normal, one plate per lot, whenever a ballistic test plate fails to meet ballistic requirements.

4.6.4.1 First level reduced testing. Testing may be reduced, upon approval by a responsible authority, to one ballistic test plate for every three lots of material of the same thickness providing the preceding five ballistic test plates of that thickness have met ballistic requirements.

4.6.4.2 Second level reduced testing. Testing may be reduced, upon approval by a responsible authority, to one ballistic test plate for every five lots of material of the same thickness providing the preceding ten ballistic test plates of that thickness have met ballistic requirements.

4.7 Rejection.

4.7.1 First article. Failure of the first article samples to meet any of the requirements of this specification indicates failure of the process.

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4.7.2 Retests. Resubmission of first article samples shall not be made until the contractor has made necessary corrections in processing the material to the satisfaction of the procuring activity.

4.7.3 Quality conformance.

4.7.3.1 Examination. A lot shall be subject to rejection for failure to meet the visual and dimensional requirements of 4.5.

4.7.3.2 Tests. A lot shall be subject to rejection for failure to meet any of the test requirements of 4.6.

4.7.3.3 Retests. At the discretion of the producer, a rejected lot may be retested as specified below.

4.7.3.3.1 Non-ballistic retests. All failed sample units from the original sample shall be excluded from the retest lot. Four new sample units shall be taken and tested in accordance with 4.6.2, as applicable. Failure of any retest specimen to meet the applicable property requirements shall be cause for rejection of the entire lot and no further retesting shall be permitted.

4.7.3.3.2 Non-ballistic retest lot. If the retest lot is accepted, the producer may, at his discretion, retest each sample unit which shows failure at one end only. Units which show failure at both ends shall not be retested and shall be rejected. Each sample unit acceptable for retesting of mechanical properties shall be retested as follows: Two test specimens shall be taken from the failed end in either or both directions, longitudinal and transverse, which showed unacceptable mechanical properties, and these specimens shall be tested in accordance with 4.6.2. If any retest shows that the mechanical properties do not conform to the requirements of Table 2 then the parent unit may be cut back at the failed end and further retested until either: (a) both ends show acceptable properties, or (b) the length of the material remaining in the parent unit is less than the minimum specified length in the acquisition documents, whichever comes first. Acceptance or rejection of the parent unit shall then be made as appropriate.

4.7.3.3.3 Ballistic retests. In the event of failure of any one or all of the ballistic test plates, the failed lot(s) may be reprocessed and two plates submitted for retest from each reprocessed lot. If a failure then occurs, the lot shall be permanently rejected.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contract the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activity within the Military Department or Defense Agency, or within the Military Department’s System Command. Packaging data retrieval is available from the managing Military Department’s or Defense Agency’s automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a geneal or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The armor specified by this specification is military unique. It is intended for use on combat vehicles. Class 1 is equivalent to material specified by previous revisions of MIL-A-46077, which is an Extra Low Interstitial (ELI) grade. Class 2 is equivalent to material specified by MIL-T-9046J, AB-1 composition. Class 3 permits a higher oxygen level which would be suitable for applications where maximum ductility and toughness are not necessary, such as appliqué. Class 4 permits compositions different from the conventional 6 Al-4 V alloy.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Class, composition and heat treat condition (annealed).
c. Products form size (thickness, width, and length) and quantity.
d. When first article sample is required (see 3.1).
e. Name of inspection agency when inspection is to be performed by other than the contractor (see 4.2).
f. Destination of ballistic test plates (see 4.2.1).
g. Additional marking, if required (see 5.1).

6.3 Stress corrosion. Certain processing procedures and service conditions may cause these titanium products to be subject to stress corrosion cracking. ARP 982C recommends practices to minimize the susceptibility to such stress corrosion cracking.

6.4 Annealing Temperature. It has been found that an annealing temperature of 1650°F provides optimum ballistic properties for Class 1 material. Reference: Effect of Annealing Temperature on the Ballistic Limit Velocity of Ti-6Al-4V ELI, Matthew Burkins and William Love, Proceedings of the 16th International Symposium on Ballistics, San Francisco, CA 23-27 September 1996.

6.5 Definitions.

6.5.1 Plate. A flat rolled product of 0.250 inch or over in thickness and over 12 inches in length with the width at least five times the thickness.

6.5.2 Capability. The words "shall be capable of" are used herein to indicate characteristics or properties required in the product but for which testing of each lot is not required. However, if such testing is performed, products not conforming to the requirements of this specification will be subject to rejection.

6.5.3 Weldable. For the purposes of this specification, a plate is defined as "weldable" if it has a composition within the limits of its class.

6.5.4 Fair impact. A fair impact is an impact resulting from the striking of the test plate by a projectile in normal flight (no yawing or tumbling) and separated from another impact or the edge of the plate, hole, crack, or spalled area by an undisturbed area of at least one caliber.

6.5.5 Witness plate. A witness plate is normally a 0.014 in thick sheet of 5052 H36 aluminum alloy (or a 0.020 in-thick sheet of 2024-T3) aluminum alloy placed 6 in behind and parallel to the test plate or other ballistic sample.

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6.5.6 Complete penetration, protection, CP(P). A protection complete penetration is a penetration in which the projectile or one or more fragments of a projectile or plate passes beyond the back of the test plate and perforates the witness plate.

6.5.7 Partial penetration, protection, PP(P). A partial penetration is any impact that is not a complete penetration.

6.5.8 V_{50} protection ballistic limit, BL(P). A BL(P) consists of an equal number of complete and partial penetrations attained by the up- and down-firing method. All BL(P)s are to be computed using the highest partial penetration velocities and the lowest complete penetration velocities. Table 5 delineates the order of priority.

Table 6. Ballistic Limits (Protection Criteria)

No. of Rounds in BL(P)	Maximum Allowable Velocity Spread (fps)	Maximum No. of Rounds To Be Used
4	60	12[1]
6	90	12
10	125[2]	as required

[1] Firing will continue until either a 4-round or 6-round BL(P) has been attained, whichever comes first in the firing order. If these occur simultaneously the 6-round BL(P) is to be reported. If, after 12 rounds have been fired and neither a 4 or 6 round BL(P) has been determined, then firing shall continue until a 10-round BL(P), having a maximum velocity spread of 125 fps, has been determined.

[2] In the event that a high partial penetration velocity occurs which is more than 125 fps above the low complete penetration velocity, then the ballistic limit consists of the 5 highest partial penetration velocities and the 5 lowest complete penetration velocities. The maximum velocity spread should be kept as small as possible without deviating from the normal up-and-down method of firing.

In the event that after following the above procedures the BL(P) is less than 25 fps above the minimum required BL(P), and a gap (high partial penetration velocity less than the low complete penetration velocity) of 25 fps or more exists in the velocities used to compute the BL(P), then one or more rounds will be fired to reduce the gap to less than 25 fps. The BL(P) will then be recomputed as before using the highest partial velocities and the lowest complete penetration velocities. The recomputed BL(P) will be the one reported on the firing record.

6.6 Ownership of ballistic test plates. First article and test plates that comply with the requirements of this specification are considered as part of the lot of titanium armor they represent, and ownership of them passes to the Government upon acceptance. Test plates that fail to comply with the requirements of this specification are considered as part of the lot they represent and remain the property of the producer just as does the rejectable lot they represent.

6.7 Relationship of mechanical properties to ballistic requirements. The minimum mechanical properties specified may not assure titanium armor plate meeting the specified ballistic requirements.

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6.8 Plate tolerances. Titanium alloy plate is normally specified to tolerances established for corrosion-resisting steel plate as specified in the applicable sections of ASTM A480 for thickness, width, length, camber, and flatness.

6.9 Other applications. For other applications not covered in this revision, e.g. those requiring the V_{50} protection ballistic limit for .50 AP, M2 at O^0 obliquity, it is suggested that reference be made to table VII or other applicable projectile protection limits outlined in the previous revision E to this specification

6.10 Metric units. When metric divisions are required, units for inch, foot, foot-pounds, feet per second, and pounds per square inch may be converted to the metric equivalent by multiplying them by the following conversion factors:

English	Multiply by	Equals	Metric SI unit
Inch	0.0254	=	meter (m)
Foot	0.3048	=	meter (m)
Pound	0.4536	=	kilogram (kg)
foot-lb	1.3558	=	joule (j)
feet/sec	0.3048	=	Meter per second (m/s)
Pounds/sq. inch	0.00689	=	megapascal (MPa)

6.11 Subject term (key word) listing.

Armor plate
Armor
Titanium
Ballistic limit
caliber .30 AP
14.5-mm API
20-mm FSP
20-mm API-T M602

Custodian:

Army – MR
Navy – OS
Air Force – ll

Preparing Activity

Army - MR

Review activities:

Army – TE
Navy – AS
Air Force – 03,16
DLA- IS

Project No 9535-0627

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REQUEST FOR BALLISTIC TEST OF TITANIUM ARMOR

FIRING RECORD:	**DATE:**	
MANUFACTURER:	**PRIME CONTRACTOR:**	
Name:	Name:	
Address:	Address:	
POC:	POC:	
Phone No:	Phone No:	
Fax No:	Fax No:	
SPECIFICATION: MIL-DTL-46077	**REVISION:** F	**AMENDMENT:**
CONTRACT NO:	TECOM PROJECT NO:	
DCAS REGION:	BALLISTIC TEST CONTRACT NO:	

TEST ITEM IDENTIFICATION:

Lot No.	Plate No.	Ordered Thickness	Alloy and Temper

PURPOSE: ___ Acceptance ___ First Article ___ Development

SAMPLE: ___ Primary ___ Retest (Firing Record No. of Failed Sample_____)

CHEMICAL ANALYSIS:

Al	V	Fe	O	C	N	H	Si				Ti
											Rem

MECHANICAL PROPERTIES:

UTS (psi):	0.2% YS (psi):	Elongation (%):

BALLISTIC TEST RESULTS:

Test	Projec-tile	Obl. (deg)	Actual Thickness (in)	Required V_{50} (fps)	Actual V_{50} (fps)	Pass/ Fail	Notes

LOTS REPRESENTED BY:	Reduced Testing	Audit Testing

Lot met (failed to meet) the ballistic requirements of specification MIL-DTL-46077F.

Government Representative	Date	Supplier Representative	Date

Figure 1. Titanium Armor Test Data Form

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Table 7. Minimum Required V_{50} Ballistic Limits (Protection Criteria).
Firing Obliquity: 0° Projectile: Caliber .30 AP M2

Thickness (in)	Required BL(P) (fps)	Thickness (in)	Required BL(P) (fps)	Thickness (in)	Required BL(P) (fps)
0.240	1,116	0.385	1,715	0.530	2,158
0.245	1,142	0.390	1,732	0.535	2,171
0.250[1]	1,167	0.395	1,749	0.540	2,185
0.255	1,192	0.400	1,766	0.545	2,199
0.260	1,216	0.405	1,783	0.550	2,212
0.265	1,239	0.410	1,799	0.555	2,225
0.270	1,263	0.415	1,815	0.560	2,239
0.275	1,286	0.420	1,832	0.565	2,252
0.280	1,308	0.425	1,848	0.570	2,265
0.285	1,330	0.430	1,864	0.575	2,278
0.290	1,352	0.435	1,879	0.580	2,291
0.295	1,373	0.440	1,895	0.585	2,304
0.300	1,394	0.445	1,910	0.590	2,317
0.305	1,415	0.450	1,926	0.595	2,330
0.310	1,436	0.455	1,941	0.600	2,342
0.315	1,456	0.460	1,956	0.605	2,355
0.320	1,476	0.465	1,971	0.610[2]	2,368
0.325	1,496	0.470	1,986	0.615	2,380
0.330	1,515	0.475	2,001	0.620	2,393
0.335	1,534	0.480	2,016	0.625	2,405
0.340	1,553	0.485	2,031	0.630	2,417
0.345	1,572	0.490	2,045	0.635	2,430
0.350	1,591	0.495	2,059	0.640	2,442
0.355	1,609	0.500	2,074	0.645	2,454
0.360	1,627	0.505	2,088	0.650	2,466
0.365	1,645	0.510	2,102	0.655	2,478
0.370	1,663	0.515	2,116	0.660	2,490
0.375	1,681	0.520	2,130	0.665	2,502
0.380	1,698	0.525	2,144	0.670	2,514

1/ Specification requirements begin with this ordered thickness.

2/ Specification requirements end with this ordered thickness.

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Table 8. Minimum Required V_{50} Ballistic Limits (Protection Criteria).
Firing Obliquity: 0° Projectile: 20-mm FSP

Thickness (in)	Required BL(P) (fps)	Thickness (in)	Required BL(P) (fps)	Thickness (in)	Required BL(P) (fps)
0.615[1]	1,904	0.765	2,292	0.915	2,817
0.620	1,913	0.770	2,308	0.920	2,836
0.625	1,922	0.775	2,324	0.925	2,854
0.630	1,932	0.780	2,341	0.930	2,872
0.635	1,942	0.785	2,357	0.935	2,890
0.640	1,952	0.790	2,374	0.940	2,908
0.645	1,963	0.795	2,391	0.945	2,927
0.650	1,973	0.800	2,408	0.950	2,945
0.655	1,985	0.805	2,425	0.955	2,963
0.660	1,996	0.810	2,442	0.960	2,981
0.665	2,008	0.815	2,459	0.965	2,999
0.670	2,020	0.820	2,477	0.970	3,017
0.675	2,032	0.825	2,494	0.975	3,035
0.680	2,045	0.830	2,512	0.980	3,053
0.685	2,057	0.835	2,529	0.985	3,071
0.690	2,070	0.840	2,547	0.990	3,089
0.695	2,084	0.845	2,565	0.995	3,107
0.700	2,097	0.850	2,582	1.000	3,124
0.705	2,111	0.855	2,600	1.005	3,142
0.710	2,125	0.860	2,618	1.010	3,160
0.715	2,139	0.865	2,636	1.015	3,177
0.720	2,154	0.870	2,654	1.020	3,195
0.725	2,168	0.875	2,672	1.025	3,212
0.730	2,183	0.880	2,690	1.030	3,230
0.735	2,198	0.885	2,708	1.035	3,247
0.740	2,213	0.890	2,727	1.040	3,264
0.745	2,229	0.895	2,745	1.045	3,281
0.750	2,244	0.900	2,763	1.050	3,299
0.755	2,260	0.905	2,781	1.055	3,316
0.760	2,276	0.910	2,799	1.060	3,332

[1/] Specification requirements begin with this ordered thickness.

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Table 8.(cont.) Minimum Required V_{50} Ballistic Limits (Protection Criteria).
Firing Obliquity: 0° Projectile: 20-mm FSP

Thickness (in)	Required BL(P) (fps)	Thickness (in)	Required BL(P) (fps)	Thickness (in)	Required BL(P) (fps)
1.065	3,349	1.215	3,808	1.365	4,166
1.070	3,366	1.220	3,821	1.370	4,176
1.075	3,383	1.225	3,835	1.375	4,186
1.080	3,399	1.230	3,848	1.380	4,196
1.085	3,416	1.235	3,861	1.385	4,207
1.090	3,432	1.240	3,874	1.390	4,217
1.095	3,448	1.245	3,887	1.395	4,227
1.100	3,465	1.250	3,900	1.400	4,236
1.105	3,481	1.255	3,913	1.405	4,246
1.110	3,497	1.260	3,926	1.410	4,256
1.115	3,513	1.265	3,938	1.415	4,266
1.120	3,528	1.270	3,950	1.420	4,275
1.125	3,544	1.275	3,963	1.425	4,285
1.130	3,560	1.280	3,975	1.430	4,294
1.135	3,575	1.285	3,987	1.435	4,304
1.140	3,591	1.290	3,999	1.440	4,313
1.145	3,606	1.295	4,011	1.445	4,322
1.150	3,621	1.300	4,022	1.450	4,331
1.155	3,636	1.305	4,034	1.455	4,341
1.160	3,651	1.310	4,046	1.460	4,350
1.165	3,666	1.315	4,057	1.465	4,359
1.170	3,681	1.320	4,068	1.470	4,368
1.175	3,695	1.325	4,080	1.475	4,377
1.180	3,710	1.330	4,091	1.480	4,386
1.185	3,724	1.335	4,102	1.485	4,395
1.190	3,738	1.340	4,113	1.490	4,404
1.195	3,752	1.345	4,123	1.495	4,413
1.200	3,766	1.350	4,134	1.500	4,422
1.205	3,780	1.355	4,145	1.505	4,431
1.210	3,794	1.360	4,155	1.510	4,440

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Table 8.(cont.) Minimum Required V_{50} Ballistic Limits (Protection Criteria).
Firing Obliquity: 0° Projectile: 20-mm FSP

Thickness (in)	Required BL(P) (fps)	Thickness (in)	Required BL(P) (fps)	Thickness (in)	Required BL(P) (fps)
1.515	4,449	1.535	4,484	1.555	4,520
1.520	4,458	1.540	4,493	1.560	4,529
1.525	4,466	1.545	4,502	1.565	4,538
1.530	4,475	1.550	4,511	1.570[2]	4,547

2/ Specification requirements end with this ordered thickness.

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Table 9. Minimum Required V_{50} Ballistic Limits (Protection Criteria)
Firing Obliquity: 0° Projectile: 14.5-mm API B32

Thickness (in)	Required BL(P) (fps)	Thickness (in)	Required BL(P) (fps)	Thickness (in)	Required BL(P) (fps)
1.575[1]	2,691	1.710	2,854	1.845	3,008
1.580	2,697	1.715	2,860	1.850	3,013
1.585	2,704	1.720	2,866	1.855	3,019
1.590	2,710	1.725	2,871	1.860	3,024
1.595	2,716	1.730	2,877	1.865	3,030
1.600	2,722	1.735	2,883	1.870	3,036
1.605	2,728	1.740	2,889	1.875	3,041
1.610	2,734	1.745	2,895	1.880	3,047
1.615	2,740	1.750	2,900	1.885	3,052
1.620	2,746	1.755	2,906	1.890	3,057
1.625	2,753	1.760	2,912	1.895	3,063
1.630	2,759	1.765	2,918	1.900	3,068
1.635	2,765	1.770	2,923	1.905	3,074
1.640	2,771	1.775	2,929	1.910	3,079
1.645	2,777	1.780	2,935	1.915	3,085
1.650	2,783	1.785	2,940	1.920	3,090
1.655	2,789	1.790	2,946	1.925	3,096
1.660	2,795	1.795	2,952	1.930	3,101
1.665	2,801	1.800	2,957	1.935	3,106
1.670	2,807	1.805	2,963	1.940	3,112
1.675	2,813	1.810	2,969	1.945[2]	3,117
1.680	2,818	1.815	2,974	1.950	3,122
1.685	2,824	1.820	2,980	1.955	3,128
1.690	2,830	1.825	2,986	1.960	3,133
1.695	2,836	1.830	2,991	1.965	3,138
1.700	2,842	1.835	2,997	1.970	3,144
1.705	2,848	1.840	3,002	1.975	3,149

1/ Specification requirements begin with this ordered thickness.
2/ Specification requirements end with this ordered thickness.

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Table 10. Minimum Required V_{50} Ballistic Limits (Protection Criteria)
Firing Obliquity: 0° Projectile: 20-mm API-T M602

Thickness (in)	Required BL(P) (fps)	Thickness (in)	Required BL(P) (fps)	Thickness (in)	Required BL(P) (fps)
1.95[1]	2,343	2.25	2,616	2.55	2,880
1.96	2,352	2.26	2,625	2.56	2,889
1.97	2,361	2.27	2,634	2.57	2,898
1.98	2,370	2.28	2,643	2.58	2,906
1.99	2,380	2.29	2,652	2.59	2,915
2.00	2,389	2.30	2,661	2.60	2,923
2.01	2,398	2.31	2,670	2.61	2,932
2.02	2,407	2.32	2,679	2.62	2,941
2.03	2,416	2.33	2,687	2.63	2,949
2.04	2,426	2.34	2,696	2.64	2,958
2.05	2,435	2.35	2,705	2.65	2,966
2.06	2,444	2.36	2,714	2.66	2,975
2.07	2,453	2.37	2,723	2.67	2,983
2.08	2,462	2.38	2,732	2.68	2,992
2.09	2,471	2.39	2,740	2.69	3,000
2.10	2,480	2.40	2,749	2.70	3,009
2.11	2,490	2.41	2,758	2.71	3,017
2.12	2,499	2.42	2,767	2.72	3,026
2.13	2,508	2.43	2,776	2.73	3,034
2.14	2,517	2.44	2,784	2.74	3,043
2.15	2,526	2.45	2,793	2.75	3,051
2.16	2,535	2.46	2,802	2.76	3,060
2.17	2,544	2.47	2,811	2.77	3,068
2.18	2,553	2.48	2,819	2.78	3,077
2.19	2,562	2.49	2,828	2.79	3,085
2.20	2,571	2.50	2,837	2.80	3,093
2.21	2,580	2.51	2,846	2.81	3,102
2.22	2,589	2.52	2,854	2.82	3,110
2.23	2,598	2.53	2,863	2.83	3,119
2.24	2,607	2.54	2,872	2.84	3,127

1/ Specification requirements begin with this ordered thickness.

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Table 10. (cont.) Minimum Required V_{50} Ballistic Limits (Protection Criteria)
Firing Obliquity: 0° Projectile: 20-mm API-T M602

Thickness (in)	Required BL(P) (fps)	Thickness (in)	Required BL(P) (fps)	Thickness (in)	Required BL(P) (fps)
2.85	3,135	3.15	3,381	3.45	3,618
2.86	3,144	3.16	3,389	3.46	3,625
2.87	3,152	3.17	3,397	3.47	3,633
2.88	3,160	3.18	3,405	3.48	3,641
2.89	3,169	3.19	3,413	3.49	3,648
2.90	3,177	3.20	3,421	3.50	3,656
2.91	3,185	3.21	3,429	3.51	3,664
2.92	3,193	3.22	3,437	3.52	3,671
2.93	3,202	3.23	3,445	3.53	3,679
2.94	3,210	3.24	3,453	3.54	3,687
2.95	3,218	3.25	3,461	3.55	3,694
2.96	3,226	3.26	3,469	3.56	3,702
2.97	3,235	3.27	3,477	3.57	3,710
2.98	3,243	3.28	3,485	3.58	3,717
2.99	3,251	3.29	3,493	3.59	3,725
3.00	3,259	3.30	3,500	3.60	3,732
3.01	3,267	3.31	3,508	3.61	3,740
3.02	3,276	3.32	3,516	3.62	3,747
3.03	3,284	3.33	3,524	3.63	3,755
3.04	3,292	3.34	3,532	3.64	3,763
3.05	3,300	3.35	3,540	3.65	3,770
3.06	3,308	3.36	3,548	3.66	3,778
3.07	3,316	3.37	3,555	3.67	3,785
3.08	3,325	3.38	3,563	3.68	3,793
3.09	3,333	3.39	3,571	3.69	3,800
3.10	3,341	3.40	3,579	3.70	3,808
3.11	3,349	3.41	3,587	3.71	3,815
3.12	3,357	3.42	3,594	3.72	3,823
3.13	3,365	3.43	3,602	3.73	3,830
3.14	3,373	3.44	3,610	3.74	3,837

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Table 10. (cont.) Minimum Required V_{50} Ballistic Limits (Protection Criteria)
Firing Obliquity: 0° Projectile: 20-mm API-T M602

Thickness (in)	Required BL(P) (fps)	Thickness (in)	Required BL(P) (fps)	Thickness (in)	Required BL(P) (fps)
3.75	3,845	3.90	3955	4.05	4,063
3.76	3,852	3.91	3,962	4.06	4,070
3.77	3,860	3.92	3,970	4.07	4,077
3.78	3,867	3.93	3,977	4.08	4,084
3.79	3,874	3.94	3,984	4.09	4,091
3.80	3,882	3.95	3,991	4.10	4,098
3.81	3,889	3.96	3,998	4.11	4,105
3.82	3,897	3.97	4,006	4.12	4,112
3.83	3,904	3.98	4,013	4.13	4,119
3.84	3,911	3.99	4,020	4.14	4,127
3.85	3,919	4.00[2]	4,027	4.15	4,134
3.86	3,926	4.01	4,034	4.16	4,141
3.87	3,933	4.02	4,042	4.17	4,148
3.88	3,940	4.03	4,049	4.18[2]	4,155
3.89	3,948	4.04	4,056	4.19	4,162
				4.20	4,168

[2/] Specification requirements end with this ordered thickness.

Note: The values above the specification requirements are for interpolation of BL(P) requirements for plates received with a plus tolerance on the nominal thickness.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter must be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-DTL-46077F	2. DOCUMENT DATE (YYMMDD) 981229

3. DOCUMENT TITLE
ARMOR PLATE, TITANIUM ALLOY, WELDABLE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2(DSN) (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME US Army Research Laboratory Weapons & Materials Research Directorate	b. TELEPHONE (Including Area Code) (1) Commercial (410) 306-0725 (2) DSN 458-0725
C. ADDRESS (Include Zip Code) ARL/WMRD ATTN: AMSRL-WM-M Aberdeen Proving Ground, MD 21005-5069	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

DD Form 1426, Oct 89 *Previous editions obsolete*

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